Catalyst-rank limits for second-order entanglement concentration

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Abstract

Exact catalysts can improve the second-order yield of pure-state entanglement concentration, but the optimal yield does not specify how the catalyst must grow. We obtain a necessary rank condition for this improvement. Let the source have a fixed nonuniform finite squared Schmidt vector, with entropy H and information variance V defined using base-two logarithms, and fix a required squared fidelity $q \in (0,1)$. For pure outputs of n source copies, with a pure catalyst returned exactly and uncorrelated, suppose that the target contains $nH + b\sqrt{n} + o(\sqrt{n})$ Bell pairs. If the catalyst rank K_n satisfies $\limsup \log_2 K_n/\sqrt{n} \le c < \infty$, we give an explicit necessary fidelity bound strictly below $e^{-b^2/(2V)}$ for every $b > 0$. Consequently, attaining the known unrestricted coefficient $\sqrt{2V \ln(1/q)}$ requires $\log_2 K_n/\sqrt{n} \to \infty$. Catalysts with $\log_2 K_n = o(\sqrt{n})$ cannot improve the unassisted coefficient. The proof combines a fidelity concentration estimate, a Gaussian limit of the majorization inequalities, and a positive tail term retained by the rank constraint.

Keywords: entanglement concentration; exact catalysis; Schmidt rank; majorization; second-order asymptotics.

1 Introduction

Entanglement concentration converts copies of a bipartite pure state into an approximation of Bell pairs by local operations and classical communication (LOCC). For a source with squared Schmidt vector p, let H be its entropy and let $V > 0$ be the variance of $-\log_2 p_I$ when I is sampled according to p. At a fixed required squared fidelity $q \in (0,1)$, the optimal number of Bell pairs without a catalyst is

$$nH + \sqrt{nV}\,\Phi^{-1}(1-q) + o(\sqrt{n}), \tag{1}$$

where Φ is the standard normal distribution function [1]. Thus the fidelity requirement affects the yield on the $\sqrt{n}$ scale even though the leading rate is H.

A catalyst supplies an auxiliary entangled state that participates in the conversion and is returned unchanged [2]. We consider exact return of a pure catalyst as a product with the pure output. In this model, Streltsov proved [3, Theorem 1] that the optimal second-order coefficient, with no restriction on catalyst size, is

$$B(q,V) = \sqrt{2V \ln(1/q)}. \tag{2}$$

In particular, when $q > 1/2$, the negative coefficient in (1) becomes positive. The optimal yield therefore leaves a resource question: how large must the catalyst be to attain this coefficient?

We answer part of that question by bounding the catalyst's Schmidt rank K_n. Every finite upper bound on $\log_2 K_n/\sqrt{n}$ forces a strictly smaller second-order coefficient than (2). It follows that an attaining sequence must satisfy

$$\frac{\log_2 K_n}{\sqrt{n}} \longrightarrow \infty. \tag{3}$$

The conclusion concerns every pure output meeting the fidelity requirement, rather than a particular concentration construction. At the other end of the scale, $\log_2 K_n = o(\sqrt{n})$ leaves the unassisted coefficient unchanged. Between these regimes we give an explicit converse bound, without asserting that it is attainable.

The need for large exact catalysts is established in other settings. Daftuar and Klimesh showed that some catalytic transformations require arbitrarily large catalyst dimension [4], and Grabowecky and Gour obtained necessary spectral and rank conditions for a specified pair of states [5]. Here the source is the prescribed tensor power $p^{\otimes n}$, the target is constrained by fidelity, and the quantity to be resolved is a growing rank on the second-order scale. Recent dimension bounds of Masajada, Philip and Streltsov apply at a fixed rate above an exponential strong-converse threshold [6]. Our target rate tends to H, so that fixed-rate dimension theorem cannot be applied directly. Their auxiliary-state and repeated-use estimates also apply to exact return; Section 5 explains the weaker rank-divergence conclusion they give at the coefficient (2).

Our proof uses the probability representation of logarithmic spectra and its compatibility with tensor products, developed by Aubrun and Nechita [7]. The rank constraint has a useful effect in this representation: after division by $\sqrt{n}$, every limiting catalyst-information variable is supported on a bounded interval. The centered source information converges to a Gaussian, whereas fidelity forces a substantial part of the output information to lie near the target level. Passing the majorization inequalities to this limit gives a convolution comparison. A positive Gaussian tail outside the catalyst's limiting support then yields the strict loss. Sections 3 and 4 develop these steps after the model and results are stated.

2 Model and rank bounds

Let $p = (p_1, \dots, p_d)$ be a fixed nonuniform probability vector with $p_i > 0$ for every i. We use base-two logarithms for information and natural logarithms in exponential expressions. Set

$$H = -\sum_{i=1}^{d} p_i \log_2 p_i, \qquad V = \sum_{i=1}^{d} p_i(-\log_2 p_i - H)^2 > 0. \tag{4}$$

All probability vectors below have finite support. They are arranged in decreasing order and padded with zeros when dimensions differ; their rank is the number of positive entries. For vectors x, y with the same total mass, $x \prec y$ means that every partial sum of x is at most the corresponding partial sum of y.

Nielsen's theorem identifies $x \prec y$ with deterministic LOCC conversion between pure states with squared Schmidt vectors x and y [8]. Accordingly, an exact catalytic conversion from n source copies to a pure output with vector s_n is described by

$$p^{\otimes n} \otimes r_n \prec s_n \otimes r_n, \qquad K_n = \operatorname{rank}(r_n), \tag{5}$$

where r_n is the squared Schmidt vector of a pure catalyst. The product on the right specifies that the catalyst is returned in the same state, without correlations with the output. No bound is imposed on classical communication.

Write $u_N = (1/N, \dots, 1/N)$ for the target spectrum. The largest squared overlap with a maximally entangled state of rank N, after alignment of the local Schmidt bases, is

$$\mathsf{F}_N(s) = \frac{1}{N}\left(\sum_{i=1}^{N}\sqrt{s_i}\right)^2. \tag{6}$$

The alignment formula follows from [9, Lemma 1]. Local basis changes are free in LOCC, so we use this aligned fidelity throughout. In particular, a protocol whose actual output has squared fidelity at least q satisfies $\mathsf{F}_N(s) \geq q$. For pure states, trace distance $D = \frac{1}{2}\|\rho - \sigma\|_1$ and squared fidelity satisfy $D = \sqrt{1-F}$; a trace-distance tolerance $\varepsilon \in (0,1)$ therefore corresponds to $q = 1 - \varepsilon^2$.

Fix $q \in (0,1)$ and nonnegative integers m_n, and put

$$N_n = 2^{m_n}, \qquad b_n = \frac{m_n - nH}{\sqrt{n}}, \qquad c_n = \frac{\log_2 K_n}{\sqrt{n}}. \tag{7}$$

We assume (5) and $\mathsf{F}_{N_n}(s_n) \geq q$ for every n. Let $\overline{\Phi} = 1 - \Phi$.

Theorem 1 (Finite rank budget)**.** *Suppose that $b_n \to b > 0$ and $\limsup_{n\to\infty} c_n \leq c$ for some finite $c \geq 0$. Then*

$$q \leq Q_V(c,b) := e^{-b^2/(2V)}\left[1 - e^{-bc/V}\overline{\Phi}\left(\frac{c}{\sqrt{V}}\right)\right] + \overline{\Phi}\left(\frac{c+b}{\sqrt{V}}\right). \tag{8}$$

For every such c and every $b > 0$,

$$Q_V(c,b) < e^{-b^2/(2V)}. \tag{9}$$

The strict inequality excludes the unrestricted coefficient at every finite rank budget. The exclusion is uniform over catalyst spectra and allowed output spectra, with the source and fidelity fixed.

Corollary 2 (Coefficient gap and necessary growth)**.** *For each fixed q, V and finite $c \geq 0$, there is $\eta = \eta(q,V,c) > 0$ such that every sequence satisfying* (5)*, the fidelity requirement, $\limsup c_n \leq c$ and $b_n \to b \in \mathbb{R}$ has*

$$b \leq B(q,V) - \eta. \tag{10}$$

Consequently, if $b_n \to B(q,V)$, then $c_n \to \infty$.

The rank measures local Schmidt-support dimension. Its base-two logarithm is the number of local support qubits up to integer rounding, so the corollary requires this number to grow faster than $\sqrt{n}$ at the unrestricted optimum. It does not give a lower bound on the catalyst's entanglement entropy or a sufficient rank for attaining that optimum.

Corollary 3 (Sub-$\sqrt{n}$ logarithmic rank)**.** *Suppose that $c_n \to 0$ and $b_n \to b \in \mathbb{R}$. Then*

$$b \leq \sqrt{V}\,\Phi^{-1}(1-q). \tag{11}$$

Thus catalysts with $\log_2 K_n = o(\sqrt{n})$ cannot improve the unassisted second-order coefficient.

Corollary 3 uses the limiting comparison directly and is stronger than the substitution $c = 0$ in (8). In particular, it applies to negative as well as positive coefficients. For positive coefficients, Figure 1 shows how the explicit bounds depend on the normalized rank budget.

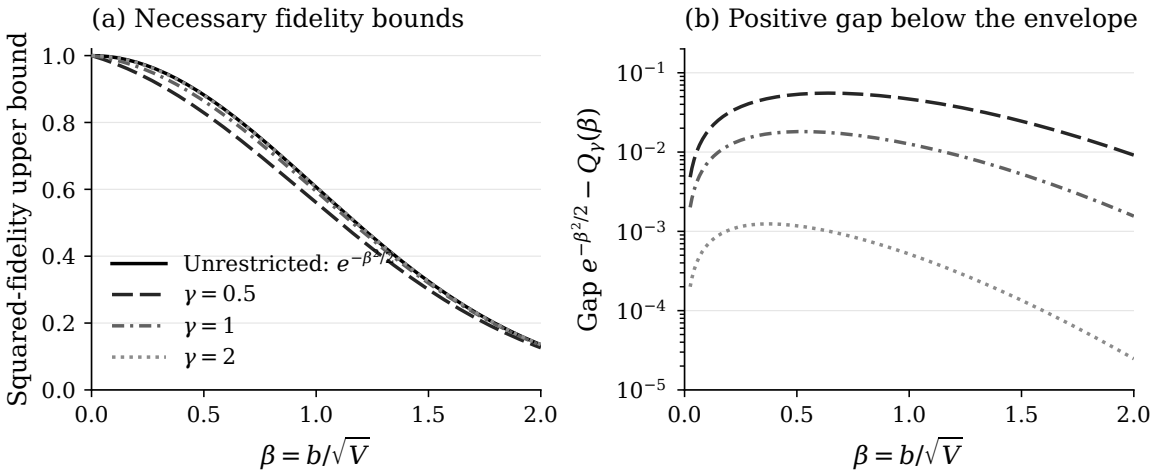


Figure 1: Necessary fidelity bounds from Theorem 1, with $\beta = b/\sqrt{V} > 0$ and $\gamma = c/\sqrt{V}$. The finite-budget curves are $Q_\gamma(\beta) = e^{-\beta^2/2}[1 - e^{-\beta\gamma}\overline{\Phi}(\gamma)] + \overline{\Phi}(\gamma + \beta)$. The unrestricted envelope is $e^{-\beta^2/2}$; the right panel displays its positive difference from each finite-budget bound. These curves are converse restrictions, not asserted achievable frontiers. The separate zero-budget conclusion is given by Corollary 3.

3 From finite spectra to a Gaussian comparison

The source and catalyst in (5) enter as a tensor product. Sampling an entry according to its own probability and taking its negative logarithm turns this product into a sum of independent random variables. We first determine what the fidelity and rank assumptions imply for the two variables on the output side. These estimates will allow us to use the central limit theorem on the source side without assuming any specific form for s_n.

3.1 The probability mass forced by fidelity

Let $I_n \sim s_n$ and define the centered output information

$$U_n = \frac{-\log_2 s_{n,I_n} - nH}{\sqrt{n}}. \tag{12}$$

Zero entries are never sampled and do not occur in this expression. The target spectrum has information m_n at every positive coordinate, corresponding to b_n after centering and scaling.

Lemma 4 (Fidelity band)**.** *For every $\delta > 0$,*

$$\mathbb{P}\{|U_n - b_n| \le \delta\} \ge \left(\sqrt{q} - 2\,2^{-\delta\sqrt{n}/2}\right)_+^2, \tag{13}$$

where $x_+ = \max\{x, 0\}$. In particular, this probability is at least $q - o(1)$.

Proof. Partition the first N_n coordinates in (6) according to whether $s_{n,i}$ is above, within, or below

$$[N_n^{-1}2^{-\delta\sqrt{n}},\ N_n^{-1}2^{\delta\sqrt{n}}].$$

There are at most $N_n 2^{-\delta\sqrt{n}}$ coordinates above this interval, since their sum is at most one. By Cauchy–Schwarz, their contribution to the root fidelity is at most

$$N_n^{-1/2}(N_n 2^{-\delta\sqrt{n}})^{1/2}\left(\sum_{\text{above}} s_{n,i}\right)^{1/2} \le 2^{-\delta\sqrt{n}/2}.$$

Each coordinate below the interval contributes at most $N_n^{-1}2^{-\delta\sqrt{n}/2}$, so all such coordinates contribute at most the same bound. If a_n is the total probability mass of the middle coordinates, Cauchy–Schwarz bounds their contribution by $\sqrt{a_n}$. The full probability in the band $|U_n - b_n| \le \delta$ is at least a_n, even when s_n has more than N_n positive entries. Hence

$$\sqrt{q} \le \sqrt{\mathsf{F}_{N_n}(s_n)} \le 2\,2^{-\delta\sqrt{n}/2} + \sqrt{\mathbb{P}\{|U_n - b_n| \le \delta\}},$$

which proves the claim. □

The lemma restricts a probability mass of approximately q but leaves the remaining mass unrestricted. This is sufficient for the converse: in the limiting majorization inequality we will allow the remainder to contribute as much as possible.

3.2 The effect of a rank budget

Independently of I_n, sample $J_n \sim r_n$ and set

$$Z_n = \frac{-\log_2 r_{n,J_n}}{\sqrt{n}} \ge 0. \tag{14}$$

A bound on rank does not give a pointwise lower bound on positive catalyst coefficients. It does, however, control the probability of sampling an exceptionally small coefficient.

Lemma 5 (Limiting catalyst support)**.** *If* $\limsup c_n \le c < \infty$*, the laws of* Z_n *are tight, and every subsequential weak limit is supported on* $[0, c]$*.*

Proof. For every $\delta > 0$,

$$\mathbb{P}\{Z_n > c + \delta\} \le K_n 2^{-(c+\delta)\sqrt{n}}. \tag{15}$$

Indeed, the event contains at most K_n coordinates, each with probability below $2^{-(c+\delta)\sqrt{n}}$. For all sufficiently large n, the limsup assumption gives $K_n \le 2^{(c+\delta/2)\sqrt{n}}$, and the right side of (15) is at most $2^{-\delta\sqrt{n}/2}$. Together with nonnegativity, these estimates give tightness and confine every weak limit to $[0, c]$. □

We may therefore pass to a subsequence on which $Z_n \Rightarrow Z$, with $0 \le Z \le c$ almost surely. The same catalyst occurs on both sides of (5). After taking logarithms, this will give the same independent additive variable Z in the source and output comparison.

3.3 Passing the majorization inequalities to the limit

For a probability vector x, define the threshold functional

$$L_x(t) = \sum_i (x_i - t)_+, \qquad t \ge 0. \tag{16}$$

Majorization implies $L_x(t) \le L_y(t)$ whenever $x \prec y$ [7, Proposition 4.2]. This threshold form is needed because majorization does not itself imply stochastic ordering of the probability-weighted logarithmic spectra [7, Remark 4.4].

Proposition 6 (Limiting convolution inequality)**.** *Suppose that* $b_n \to b \in \mathbb{R}$ *and* $\limsup c_n \le c < \infty$*. For every subsequential limit* Z *from Lemma 5, let* $G \sim N(0, V)$ *be independent of* Z*. Then, for every* $a \in \mathbb{R}$*,*

$$\mathbb{P}\{G + Z \le a\} \le 1 - q + q\mathbb{P}\{Z \le a - b\}. \tag{17}$$

Proof. Work along a subsequence on which $Z_n \Rightarrow Z$, and retain the index n. For any vector $x^{(n)}$, sample an entry with its own probability and write

$$W_n = \frac{-\log_2 x_I^{(n)} - nH}{\sqrt{n}}, \qquad t_n(a) = 2^{-nH - a\sqrt{n}}.$$

Expanding (16) with these probabilities gives

$$L_{x^{(n)}}(t_n(a)) = \mathbb{E}\left[(1 - 2^{-\sqrt{n}(a - W_n)})\mathbf{1}_{\{W_n < a\}}\right]. \tag{18}$$

Consequently, for every $\rho > 0$,

$$(1 - 2^{-\rho\sqrt{n}})\mathbb{P}\{W_n \le a - \rho\} \le L_{x^{(n)}}(t_n(a)) \le \mathbb{P}\{W_n < a\}. \tag{19}$$

For the source vector $p^{\otimes n} \otimes r_n$, the variable W_n is $X_n + Z_n$, where

$$X_n = \frac{\sum_{\ell=1}^{n}(-\log_2 p_{A_\ell}) - nH}{\sqrt{n}}, \qquad A_1, \dots, A_n \text{ independent with law } p.$$

The central limit theorem gives $X_n \Rightarrow G$. Independence of the source and catalyst implies $X_n + Z_n \Rightarrow G + Z$. Since convolution with a nondegenerate Gaussian has a continuous distribution function, (19), followed by $\rho \downarrow 0$, yields

$$L_{p^{\otimes n} \otimes r_n}(t_n(a)) \longrightarrow \mathbb{P}\{G + Z \le a\}. \tag{20}$$

For the output vector $s_n \otimes r_n$, the corresponding variable is $U_n + Z_n$. Fix $\delta > 0$ and write

$$\alpha_n = \mathbb{P}\{|U_n - b_n| \le \delta\}, \qquad q_{n,\delta} = \left(\sqrt{q} - 2\,2^{-\delta\sqrt{n}/2}\right)_+^2.$$

Lemma 4 gives $\alpha_n \ge q_{n,\delta}$ and $q_{n,\delta} \to q$. On the band, $U_n \ge b_n - \delta$; off the band we use the upper bound one for a conditional probability. Independence therefore gives

$$\begin{aligned} \mathbb{P}\{U_n + Z_n < a\} &\le 1 - \alpha_n + \alpha_n \mathbb{P}\{Z_n < a - b_n + \delta\} \\ &\le 1 - q_{n,\delta} + q_{n,\delta}\mathbb{P}\{Z_n < a - b_n + \delta\}. \end{aligned} \tag{21}$$

The second inequality uses the fact that $1 - x + xv$ decreases with x when $0 \le v \le 1$.

Combining (5), (19), (20) and (21) reduces the desired limit to the last catalyst probability. To include possible atoms of Z, fix an additional slack $\xi > 0$. Eventually

$$a - b_n + \delta < a - b + \delta + \xi.$$

Portmanteau applied to the closed half-line then gives

$$\limsup_n \mathbb{P}\{Z_n < a - b_n + \delta\} \le \mathbb{P}\{Z \le a - b + \delta + \xi\}.$$

First let $\xi \downarrow 0$, and then $\delta \downarrow 0$. Continuity from above of probability yields (17) at every a, without a continuity assumption on the law of Z. □

The comparison retains the full mass of the Gaussian source fluctuation, while only the mass q forced by fidelity is retained from the output. Exact uncorrelated return makes it possible to use the same independent catalyst distribution in both terms. The proof has not required a weak limit for the remaining output information.

4 The positive tail beyond the catalyst support

Proposition 6 turns the rank problem into a statement about a bounded random variable Z and a Gaussian G. Exponential integration of their survival probabilities first gives the unrestricted Gaussian expression. Keeping the contribution beyond c, where Z has no mass, makes that bound strict.

Lemma 7 (Tail estimate)**.** *Let $0 \le Z \le c$ almost surely, let $G \sim N(0, V)$ be independent of Z, and suppose that (17) holds for some $q \in (0,1)$ and $b \in \mathbb{R}$. For every $\theta > 0$,*

$$e^{-\theta b + V\theta^2/2} - q \ge \theta e^{-\theta c} \int_c^\infty e^{\theta u} \overline{\Phi}\Big(\frac{u+b}{\sqrt{V}}\Big)\, du > 0. \tag{22}$$

Proof. Taking complements in (17) at $a = u + b$ gives

$$D(u) := \mathbb{P}\{G + Z > u + b\} - q\mathbb{P}\{Z > u\} \ge 0. \tag{23}$$

For a real random variable Y with finite positive exponential moment, Tonelli's theorem gives

$$\int_{\mathbb{R}} e^{\theta u} \mathbb{P}\{Y > u\}\, du = \mathbb{E} \int_{-\infty}^{Y} e^{\theta u}\, du = \frac{\mathbb{E}e^{\theta Y}}{\theta}. \tag{24}$$

Apply this identity to $Y = G + Z - b$ and to $Y = Z$. Both integrals are finite, so their difference is

$$\int_{\mathbb{R}} e^{\theta u} D(u)\, du = \frac{\mathbb{E}e^{\theta Z}}{\theta} \left(e^{-\theta b + V\theta^2/2} - q\right). \tag{25}$$

Here independence factors the exponential moment, and the Gaussian moment is $\mathbb{E}e^{\theta G} = e^{V\theta^2/2}$.

For $u > c$, the second probability in (23) vanishes. Since $Z \ge 0$, the first is bounded below by the Gaussian tail:

$$D(u) \ge \overline{\Phi}\Big(\frac{u+b}{\sqrt{V}}\Big), \qquad u > c.$$

Nonnegativity of D on the rest of the line and (25) imply

$$e^{-\theta b + V\theta^2/2} - q \ge \frac{\theta}{\mathbb{E}e^{\theta Z}} \int_c^\infty e^{\theta u} \overline{\Phi}\Big(\frac{u+b}{\sqrt{V}}\Big)\, du.$$

The integral is finite and strictly positive. Finally, $\mathbb{E}e^{\theta Z} \le e^{\theta c}$ proves (22). □

Proof of Theorem 1. Use Lemma 5 to select a limit Z and Proposition 6 to obtain (17). Put $A_\theta = e^{-\theta b + V\theta^2/2}$. Integration by parts, with a vanishing boundary term at infinity, gives

$$\begin{aligned} &\theta e^{-\theta c} \int_c^\infty e^{\theta u} \overline{\Phi}\Big(\frac{u+b}{\sqrt{V}}\Big)\, du \\ &\quad = -\overline{\Phi}\Big(\frac{c+b}{\sqrt{V}}\Big) + \frac{e^{-\theta c}}{\sqrt{V}} \int_c^\infty e^{\theta u} \phi\Big(\frac{u+b}{\sqrt{V}}\Big)\, du \\ &\quad = A_\theta e^{-\theta c} \overline{\Phi}\Big(\frac{c+b-\theta V}{\sqrt{V}}\Big) - \overline{\Phi}\Big(\frac{c+b}{\sqrt{V}}\Big), \end{aligned} \tag{26}$$

where ϕ is the standard normal density. The last equality follows by completing the square,

$$\theta u - \frac{(u+b)^2}{2V} = -\theta b + \frac{V\theta^2}{2} - \frac{(u+b-\theta V)^2}{2V}.$$

The Gaussian expression A_θ is minimized at $\theta = b/V$, which is positive because $b > 0$. Use this choice in Lemma 7; substitution of (26) then gives (8). The subtracted integral in (22) is strictly positive, which also proves (9). □

Proof of Corollary 2. Write $B = B(q,V) > 0$. Since $e^{-B^2/(2V)} = q$, Theorem 1 gives $Q_V(c,B) < q$. Continuity in b provides $\eta \in (0,B)$ such that $Q_V(c,b) < q$ whenever $B-\eta < b \le B$. For $b \ge B$, (9) gives $Q_V(c,b) < e^{-b^2/(2V)} \le q$. All these coefficients are excluded by (8); coefficients $b \le 0$ already satisfy (10).

Now suppose $b_n \to B$ but c_n does not tend to infinity. There are a finite c and an infinite subsequence with $c_n \le c$. The proof of Theorem 1 applies along that subsequence and gives $q \le Q_V(c,B) < q$, a contradiction. □

Proof of Corollary 3. When $c_n \to 0$, Lemma 5 gives $Z_n \Rightarrow 0$. Proposition 6, which allows every finite real b, therefore implies

$$\Phi(a/\sqrt{V}) \le 1-q \qquad (a<b).$$

Letting $a \uparrow b$ and applying Φ^{-1} proves (11). The bound is attained at the level of the second-order coefficient by ordinary concentration (1), with no catalyst. □

5 Comparison with repeated catalyst use

Exact product return permits the same catalyst to be used on successive, independent blocks. The auxiliary-state and repeated-use bounds of Masajada, Philip and Streltsov [6, Lemmas 13–14] consequently imply, in the present pure-catalyst setting,

$$q^k \le K_n F^{\mathrm{un}}(kn, km_n), \qquad k \ge 1, \tag{27}$$

where $F^{\mathrm{un}}(N,m)$ is the largest unassisted squared fidelity obtainable from N source copies with m Bell pairs. One may see the factor K_n directly: the pure catalyst $|\chi_{r_n}\rangle = \sum_j \sqrt{r_{n,j}}|jj\rangle$ satisfies

$$|\chi_{r_n}\rangle\langle\chi_{r_n}| \le K_n \sum_j r_{n,j}|jj\rangle\langle jj|,$$

by Cauchy–Schwarz. The state on the right after division by K_n is separable and can be prepared by LOCC. Applying the repeated protocol, discarding the catalyst, and taking the overlap with the joint target preserves the operator inequality. The separable input therefore gives an unassisted protocol, while exact product return on the original input leaves k independent outputs with joint squared fidelity at least q^k. This proves (27).

For a pure target and pure input, the optimum defining F^{un} is attained by a deterministic pure output [9, Lemma 2]. For fixed k, the repeated target has coefficient $(km_n - knH)/\sqrt{kn} = \sqrt{k}\,b_n \to b\sqrt{k}$. Thus the ordinary second-order concentration formula [1, Eq. (177)], with root fidelity squared, gives

$$F^{\mathrm{un}}(kn, km_n) \longrightarrow \overline{\Phi}\left(\frac{b\sqrt{k}}{\sqrt{V}}\right). \tag{28}$$

At $b = B(q,V)$, the Gaussian tail asymptotic yields

$$\frac{q^k}{\overline{\Phi}(B\sqrt{k/V})} \sim \frac{B}{\sqrt{V}}\sqrt{2\pi k} \qquad (k\to\infty). \tag{29}$$

Equations (27)–(29), taking $n \to \infty$ for fixed k before increasing k, show that $K_n \to \infty$ at the unrestricted coefficient.

This argument gives no rate for that divergence. Substituting a growing $k = k_n$ in (28) would require a uniform estimate not supplied by the fixed-k limit; in particular, $b_n - B = o(1)$ may decay arbitrarily slowly. The bounded-support comparison of Proposition 6 retains the scale of $\log_2 K_n$ before taking the limit and leads to the stronger conclusion (3).

6 Conclusion

For exact pure catalysts returned without correlations, a finite budget on $\log_2 K_n/\sqrt{n}$ imposes a strict loss below the optimal second-order concentration coefficient. The loss follows from a Gaussian tail outside the limiting catalyst-information support and holds for every pure output satisfying the fixed fidelity requirement. A vanishing budget on this scale recovers the ordinary concentration coefficient.

The resulting rank condition is necessary. Determining the optimal coefficient at a fixed positive budget, or constructing catalysts whose rank growth approaches the necessary scale, requires additional arguments. The proof also leaves open corresponding resource bounds for mixed outputs, mixed catalysts, or return conditions allowing error or correlations.

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Data and code. No experimental data are used. Figure 1 is generated directly from (8); its Python source accompanies the manuscript sources. The plotted evaluations are illustrative and are not used in the proof.

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